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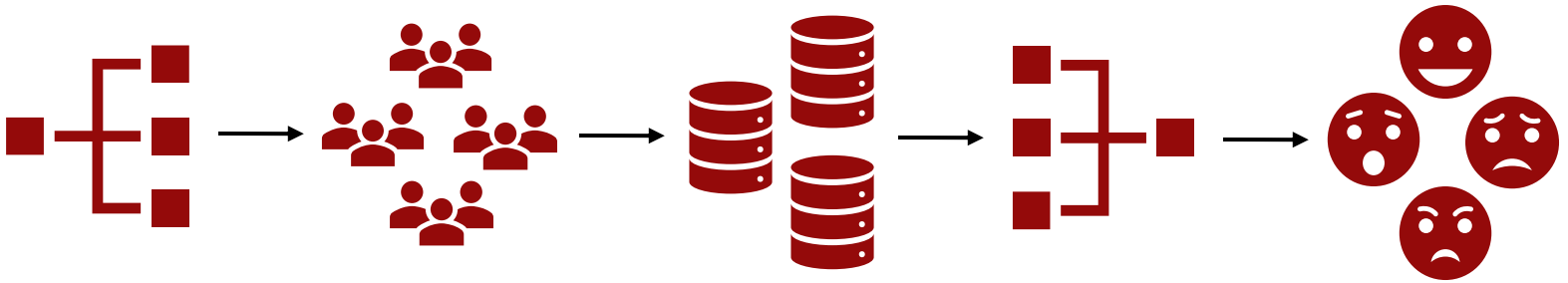


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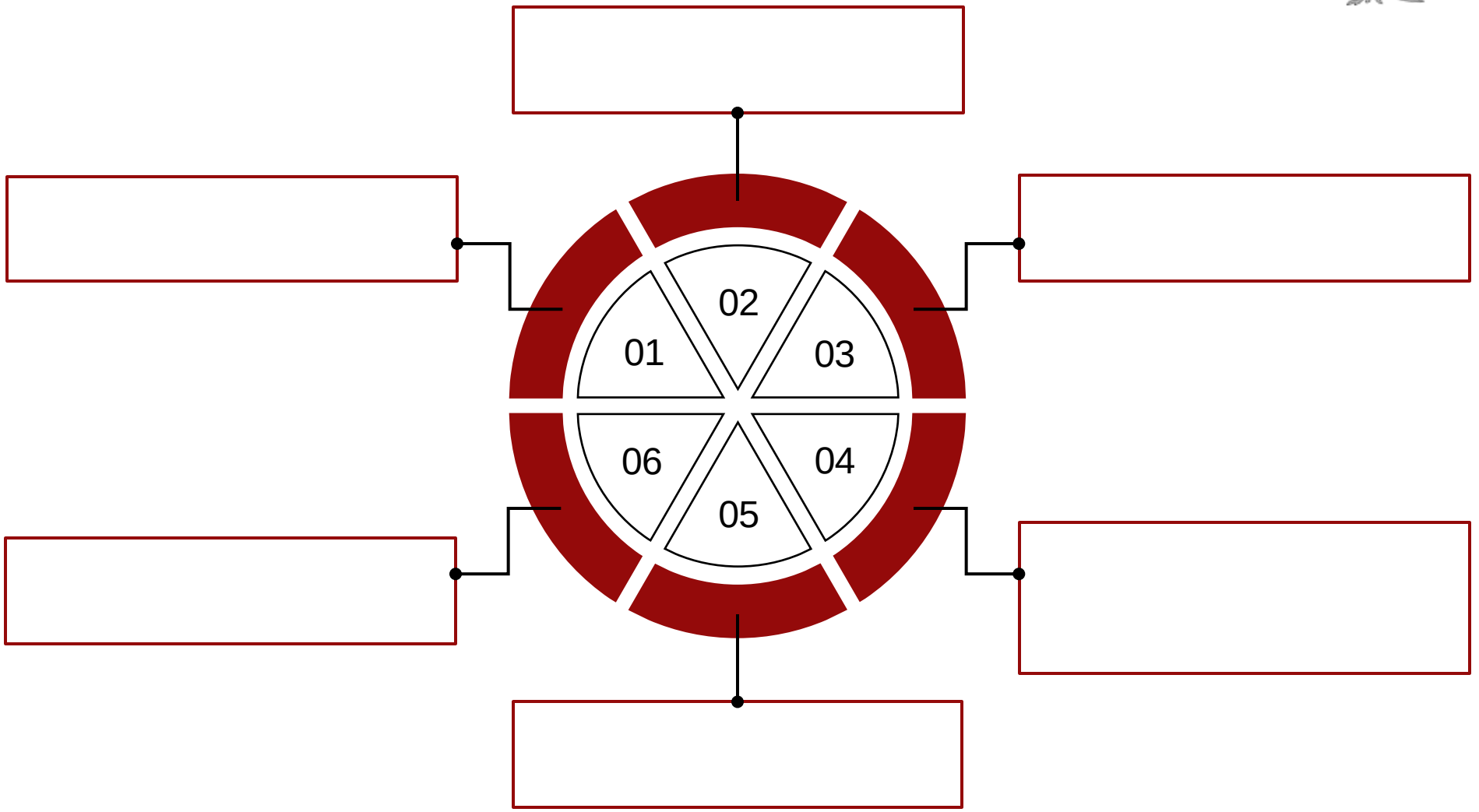
01 02 03

04 05 06

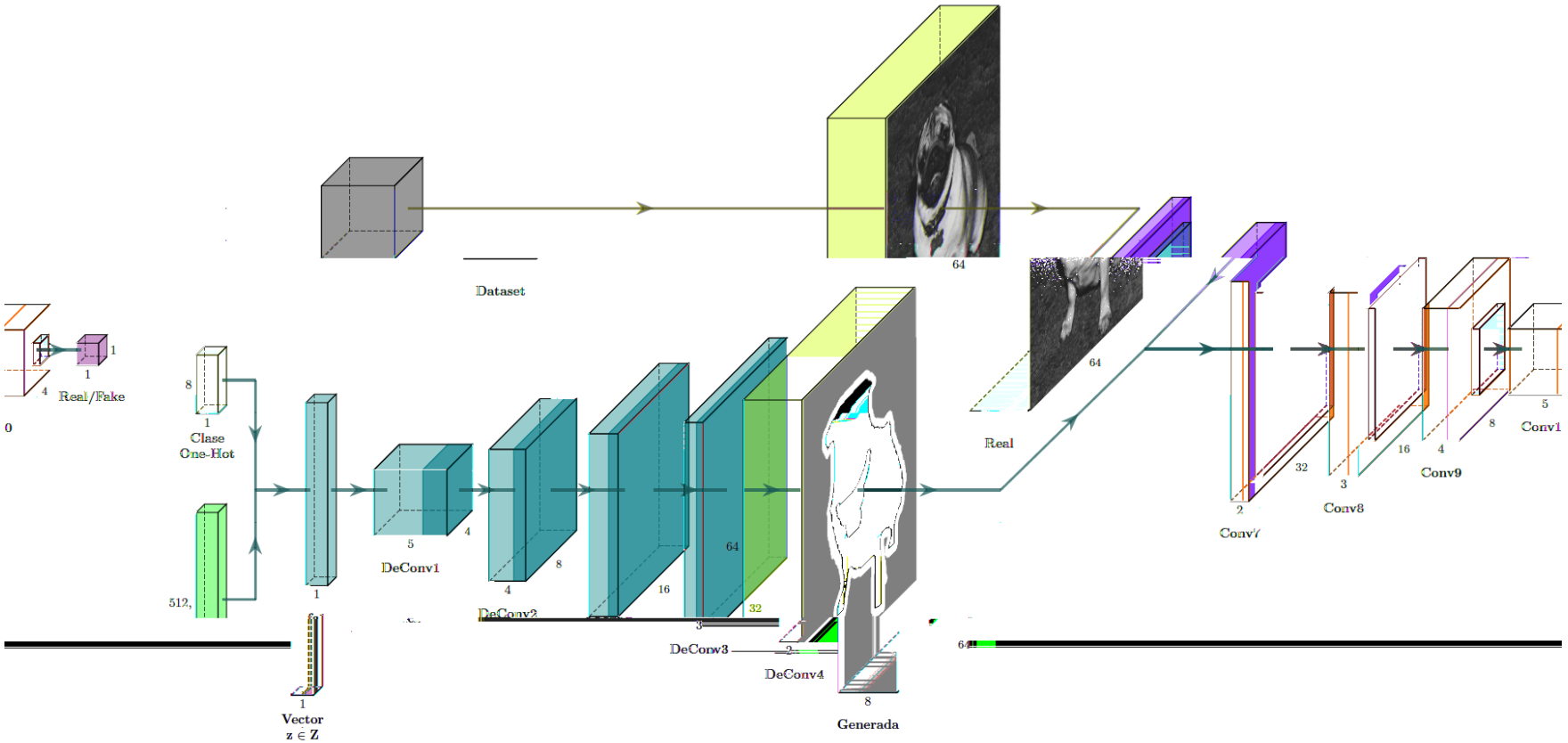
01



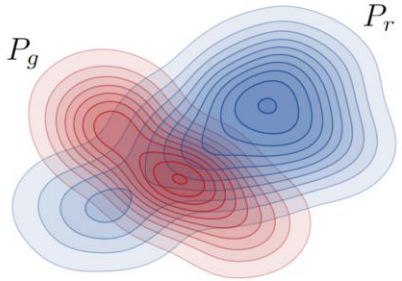
01

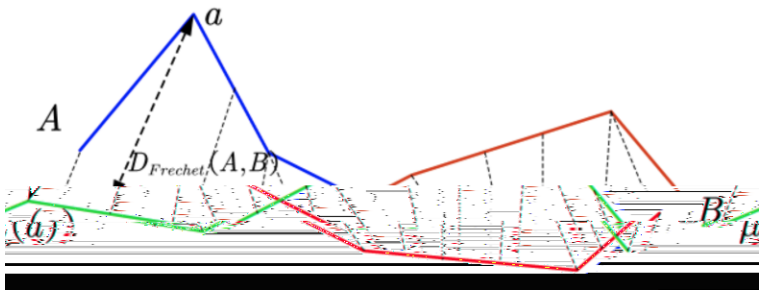


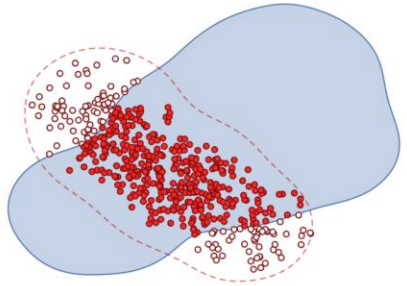
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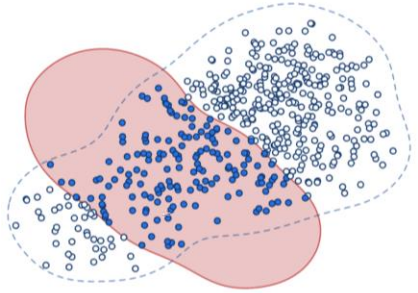

Dataset
64
Real/Fake
Clase
One-Hot
DeConv1
DeConv2
DeConv3
DeConv4
Vector
z ∈ Z
Generada
Real
Conv7
Conv8
Conv9
Conv1

02







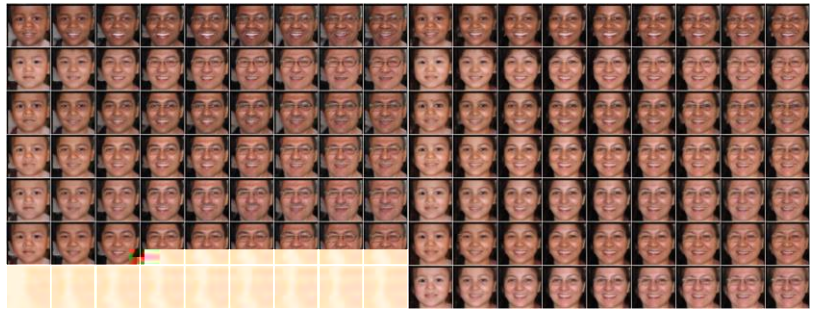


R

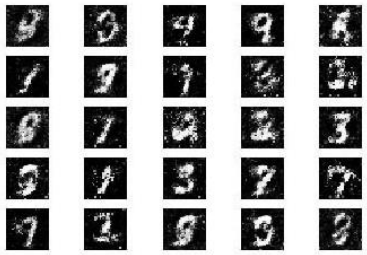
02







R

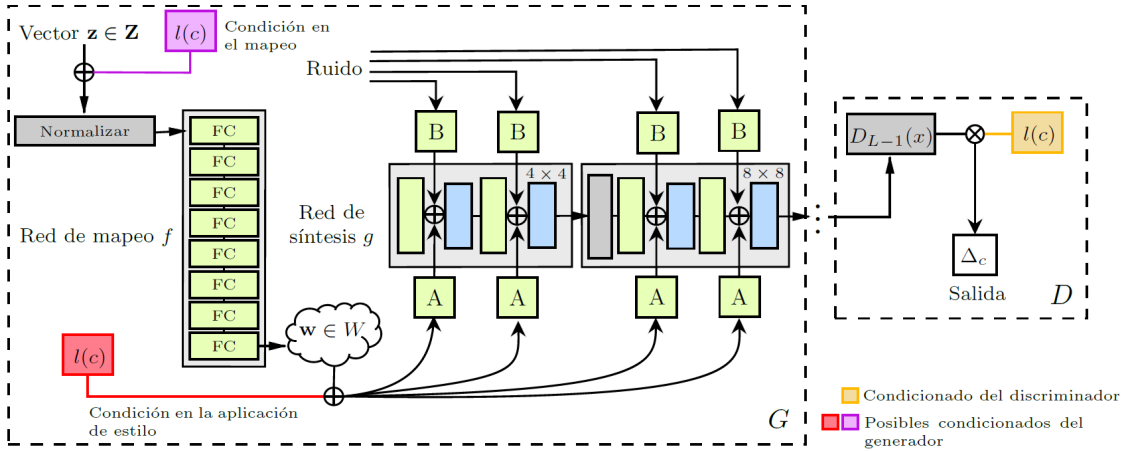




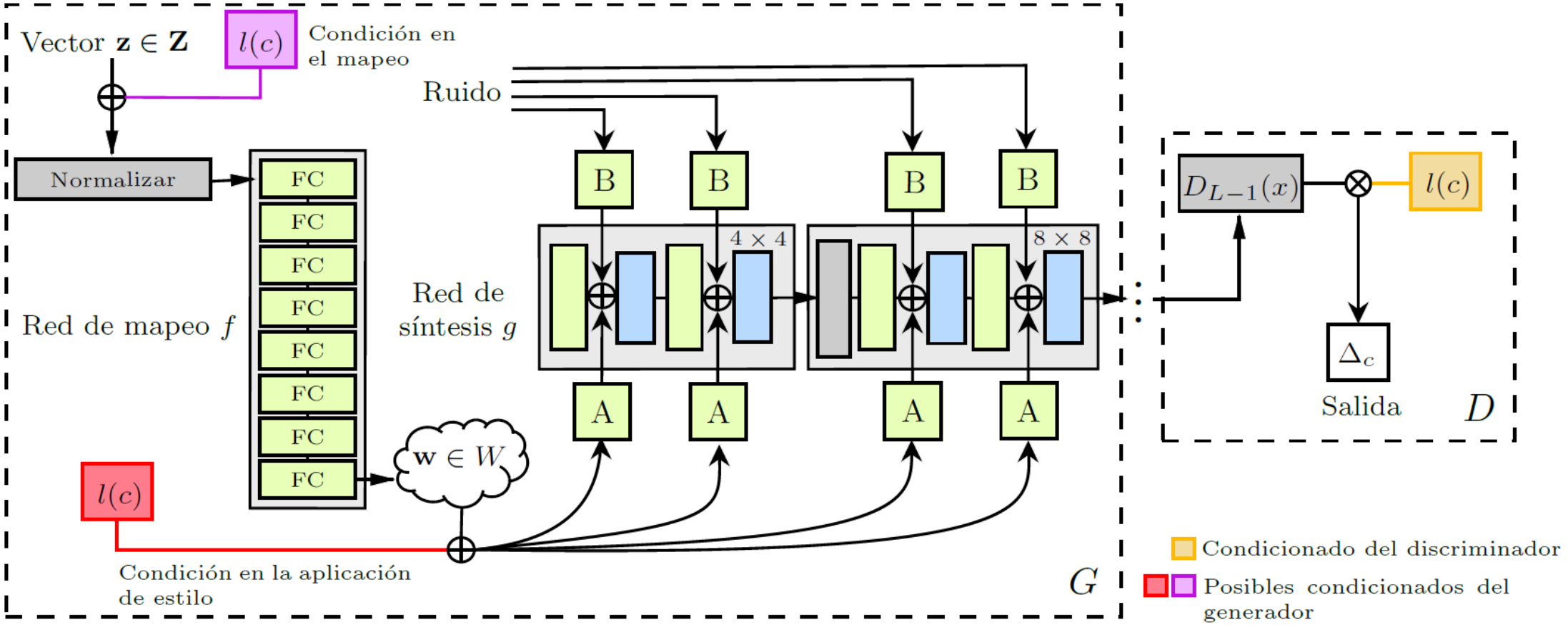


03

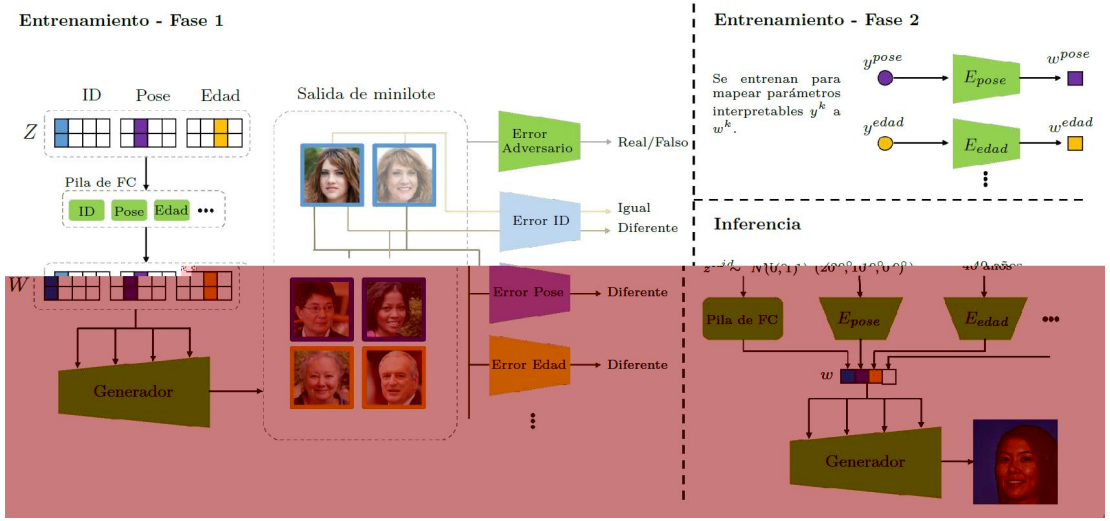
- mapeo
- estilo
- salida



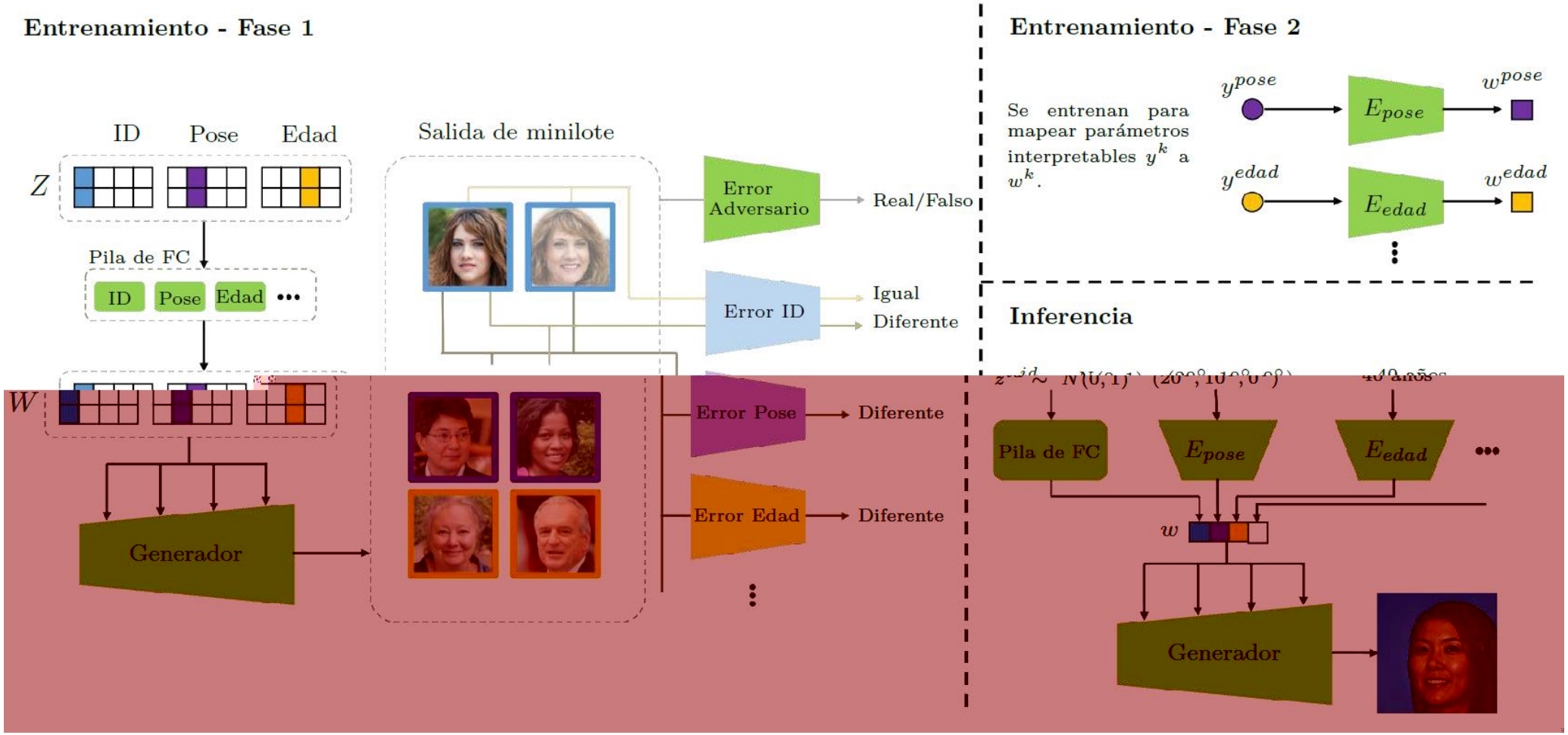
03


Vector $\mathbf{z} \in \mathbf{Z}$
$l(c)$
Condición en el mapeo
Ruido
Normalizar
FC
FC
FC
FC
FC
FC
FC
FC
Red de mapeo f
Red de síntesis g
B
B
B
B
4 × 4
8 × 8
A
A
A
A
$\mathbf{w} \in W$
$l(c)$
Condición en la aplicación de estilo
G
$D_{L-1}(x)$
$l(c)$
Δ_c
Salida
D
Condicionado del discriminador
Posibles condicionados del generador

03


Entrenamiento - Fase 1
ID
Pose
Edad
Z
Pila de FC
ID
Pose
Edad
W
Generador
Salida de minilote
Error Adversario
Real/Falso
Error ID
Igual
Diferente
Error Pose
Diferente
Error Edad
Diferente
Entrenamiento - Fase 2
Se entrenan para mapear parámetros interpretables y^k a w^k.
y^{pose}
E_{pose}
w^{pose}
y^{edad}
E_{edad}
w^{edad}
Inferencia
Pila de FC
E_{pose}
E_{edad}
w
Generador

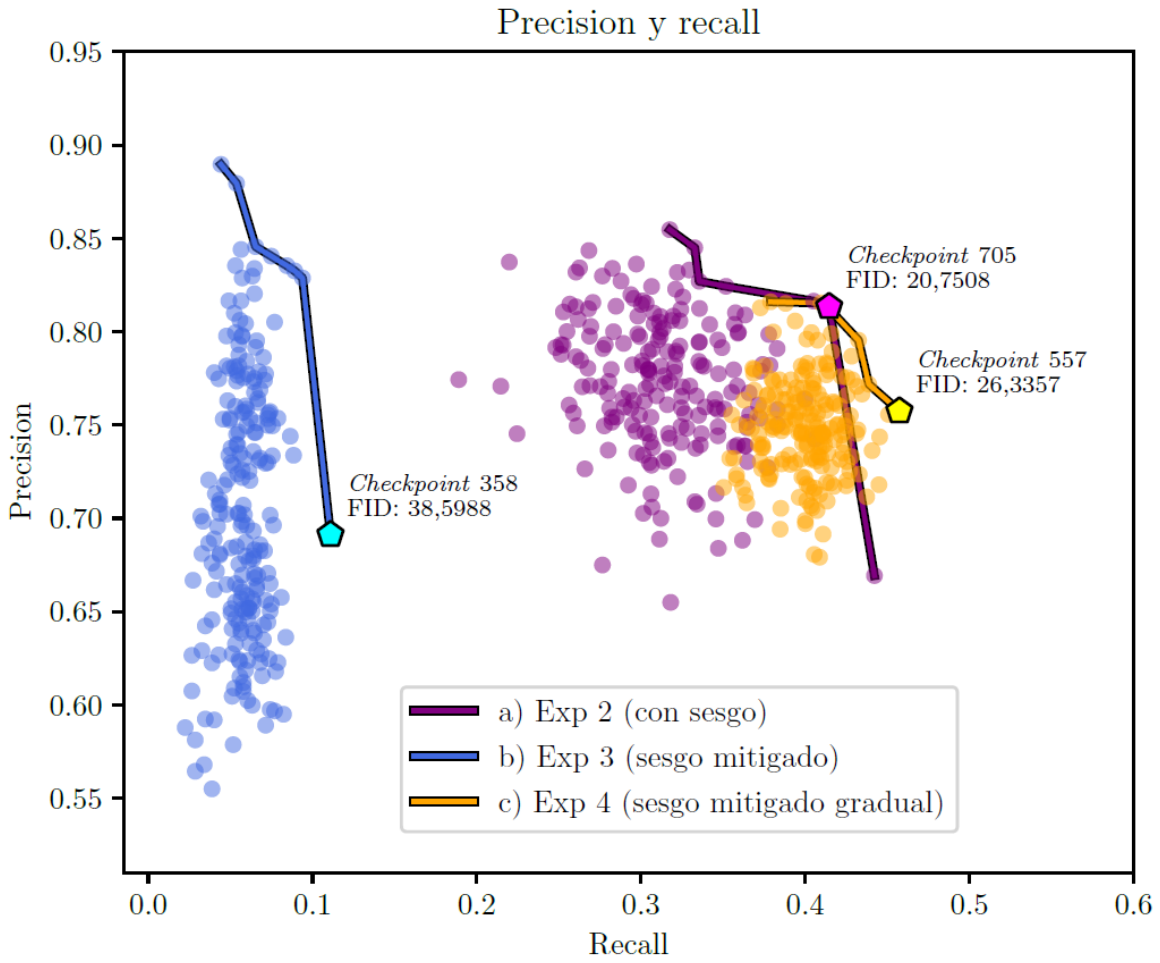
03


Entrenamiento - Fase 1
ID
Pose
Edad
Z
Pila de FC
ID
Pose
Edad
W
Generador
Salida de minilote
Error Adversario
Real/Falso
Error ID
Igual
Diferente
Error Pose
Diferente
Error Edad
Diferente
Entrenamiento - Fase 2
Se entrenan para mapear parámetros interpretables y^k a w^k.
y^{pose}
E_{pose}
w^{pose}
y^{edad}
E_{edad}
w^{edad}
Inferencia
Pila de FC
E_{pose}
E_{edad}
w
Generador

04

-
  -
  -
  -
  -
  - ×
-
  -
  -
  -
  -
  - ×
-
  -
  -
  -
  -
  - ×
-
  -
  -
  -
  -
  - ×

04


Precision y recall
Precision
Recall
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60
0.55
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Checkpoint 705
FID: 20,7508
Checkpoint 557
FID: 26,3357
Checkpoint 358
FID: 38,5988
a) Exp 2 (con sesgo)
b) Exp 3 (sesgo mitigado)
c) Exp 4 (sesgo mitigado gradual)

04

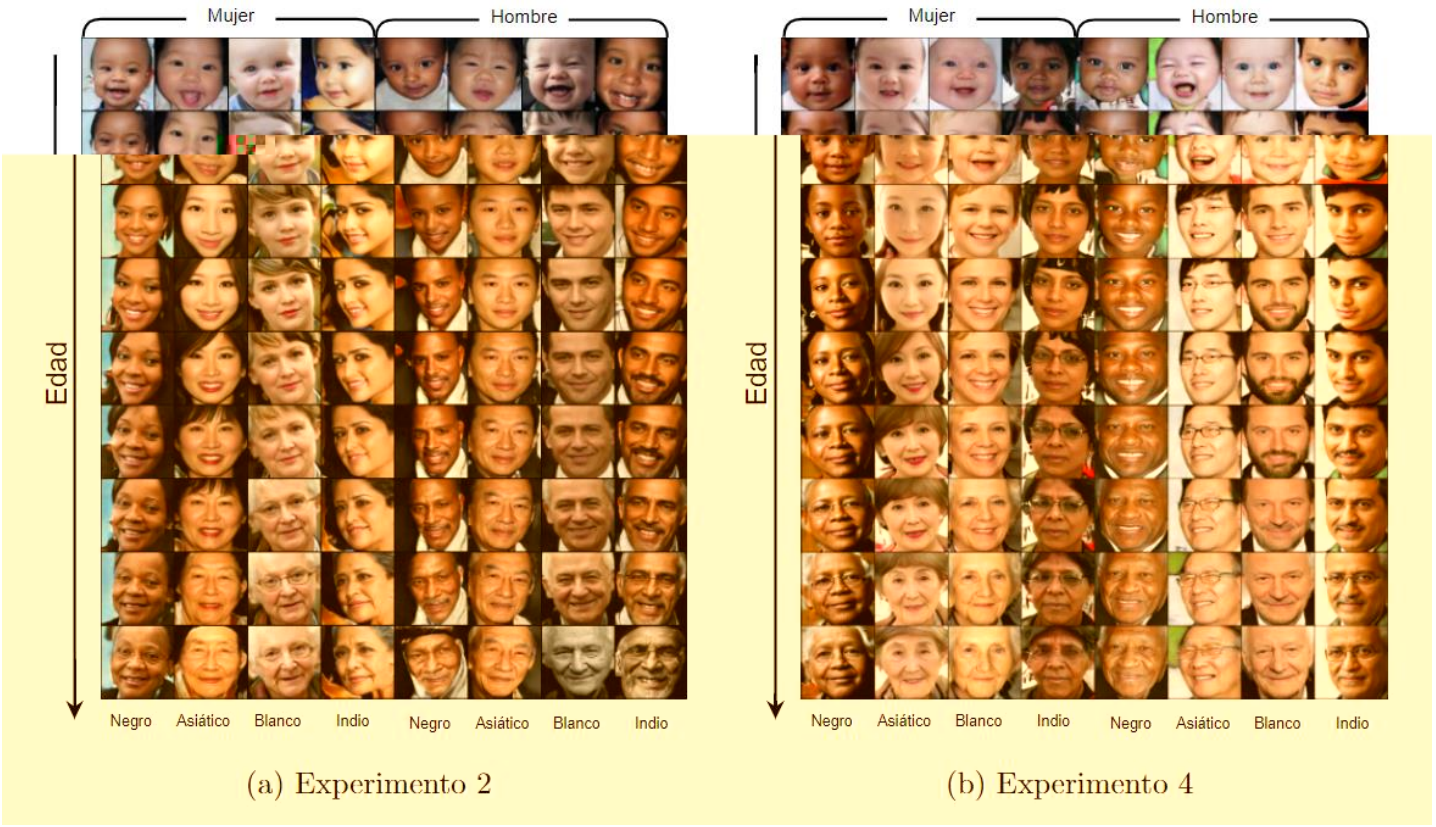


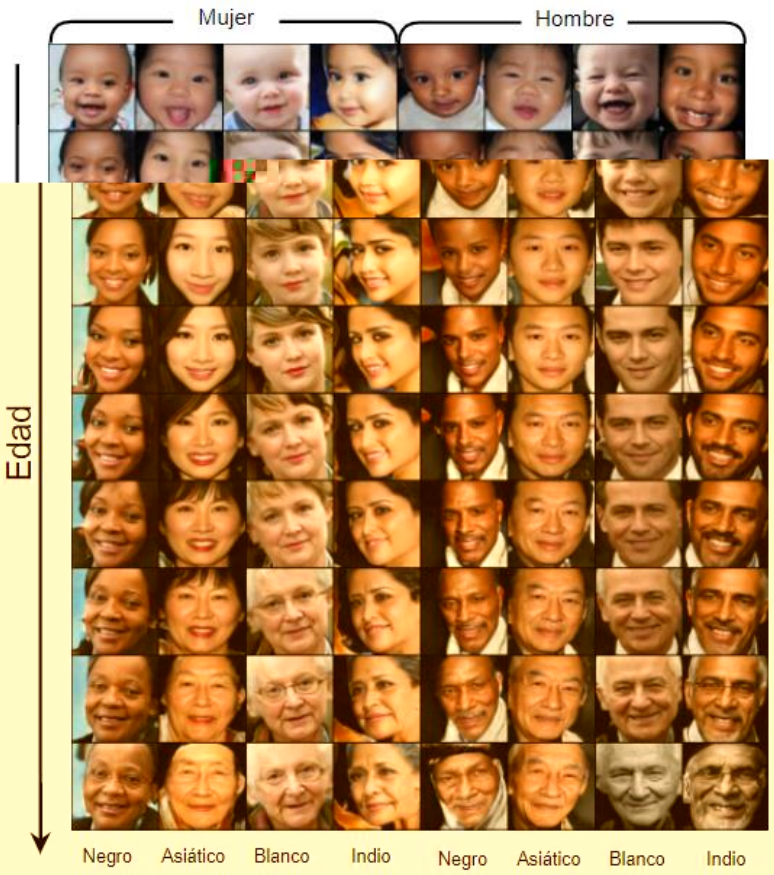
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Experimento	FID ↓	Precision ↑	Recall ↑	F_1 ↑
2 (con sesgo)	**20,7508 ± 0,5631**	**0,8134 ± 0,0052**	0,4146 ± 0,0072	0,5492 ± 0,0057
3 (sin sesgo)	38,5988 ± 0,8482	0,6912 ± 0,0099	0,1109 ± 0,0032	0,1912 ± 0,0051
4 (con medio sesgo)	26,3357 ± 0,2772	0,7574 ± 0,0058	**0,4574 ± 0,0087**	**0,5704 ± 0,0081**

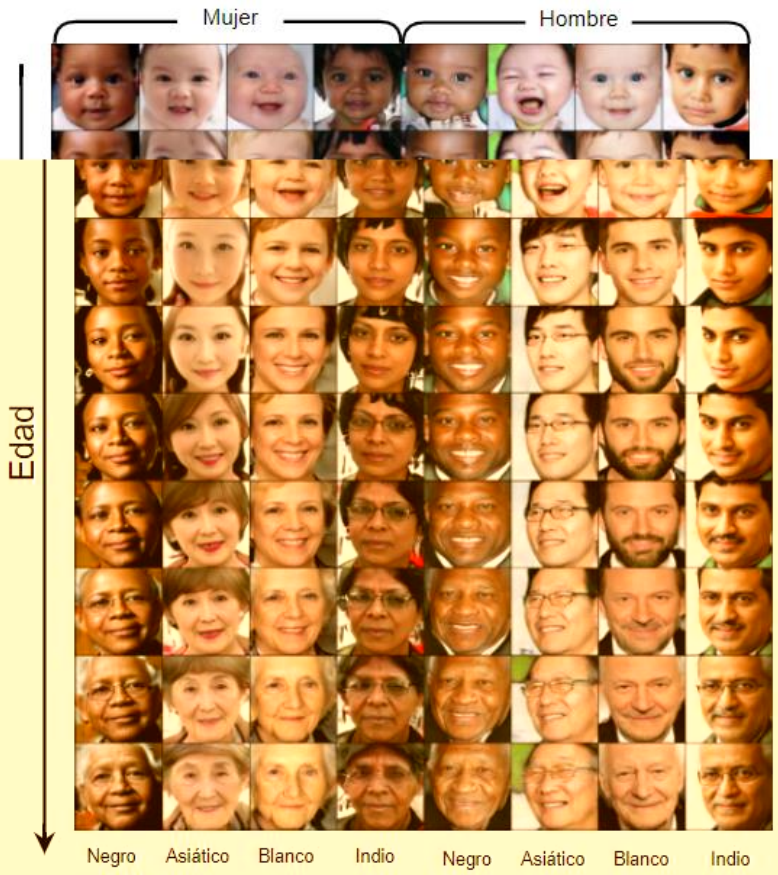
04

-
  -
    - R
  -
    - R
-
  - R


Mujer
Hombre
Edad
Negro
Asiático
Blanco
Indio
Negro
Asiático
Blanco
Indio
(a) Experimento 2
Mujer
Hombre
Edad
Negro
Asiático
Blanco
Indio
Negro
Asiático
Blanco
Indio
(b) Experimento 4



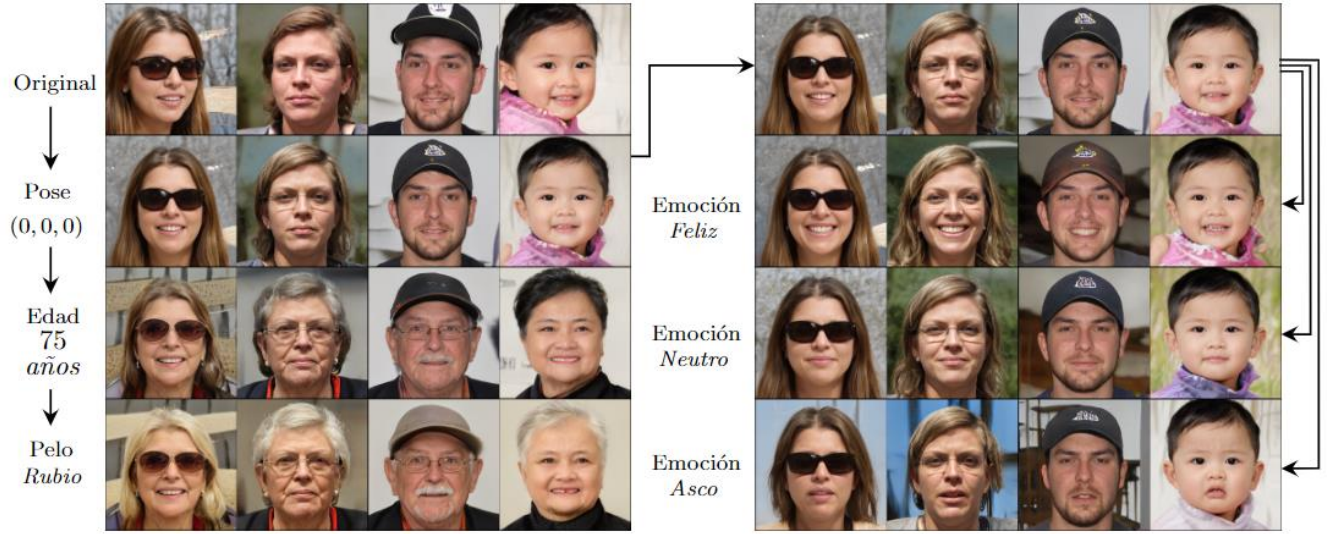
(a) Experimento 2



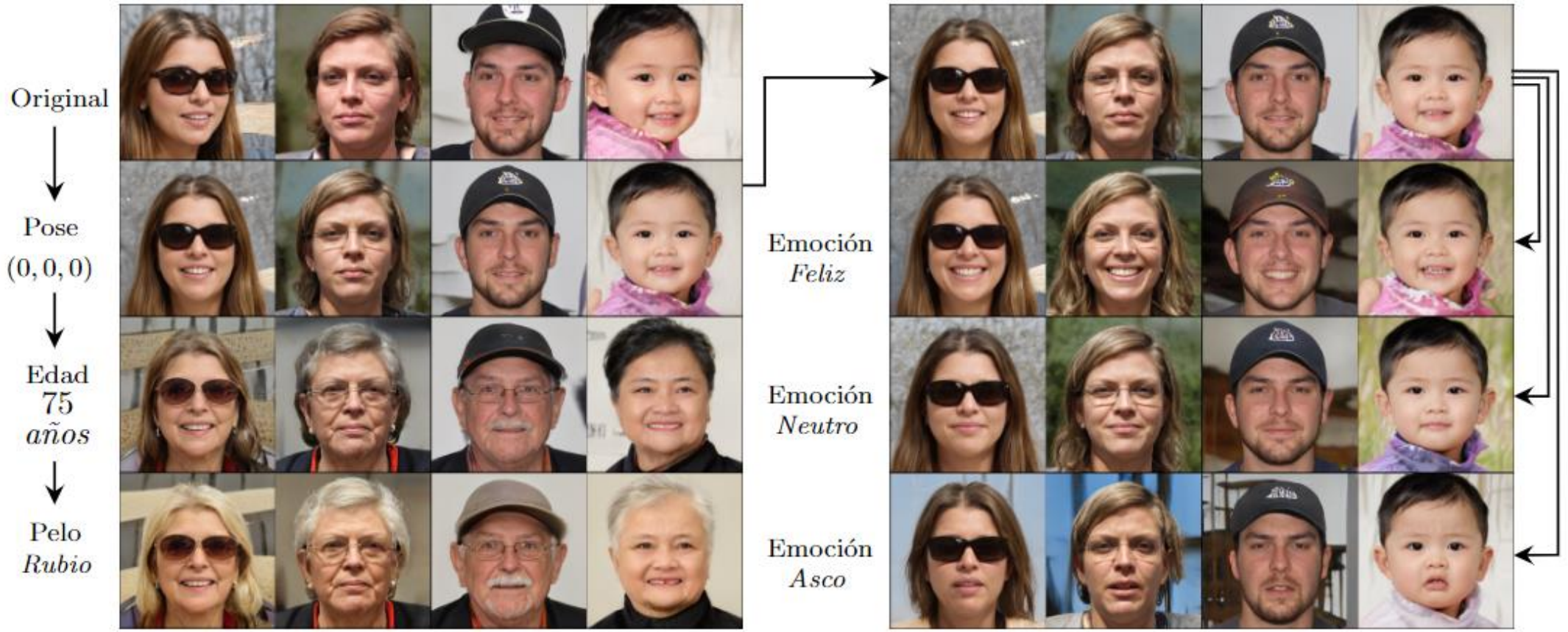
(b) Experimento 4

04

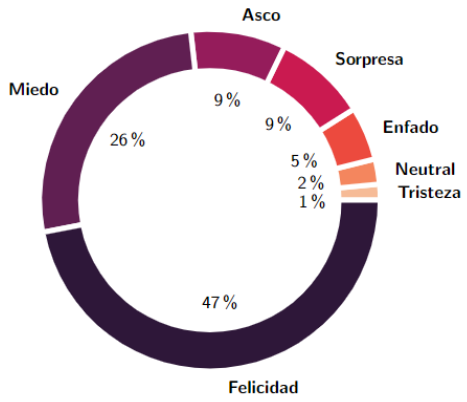
-
  -
-
  - R


Original
Pose
(0, 0, 0)
Edad
75
años
Pelo
Rubio
Emoción
Feliz
Emoción
Neutro
Emoción
Asco

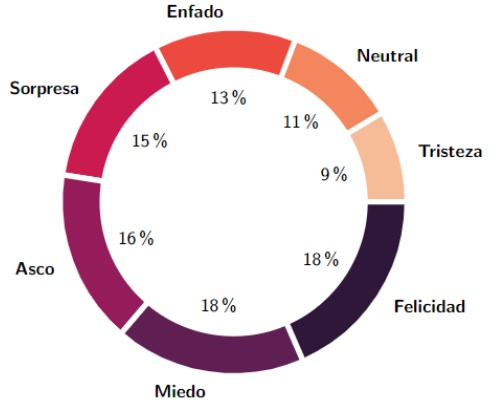
04


Original
Pose
(0, 0, 0)
Edad
75
años
Pelo
Rubio
Emoción
Feliz
Emoción
Neutro
Emoción
Asco

05



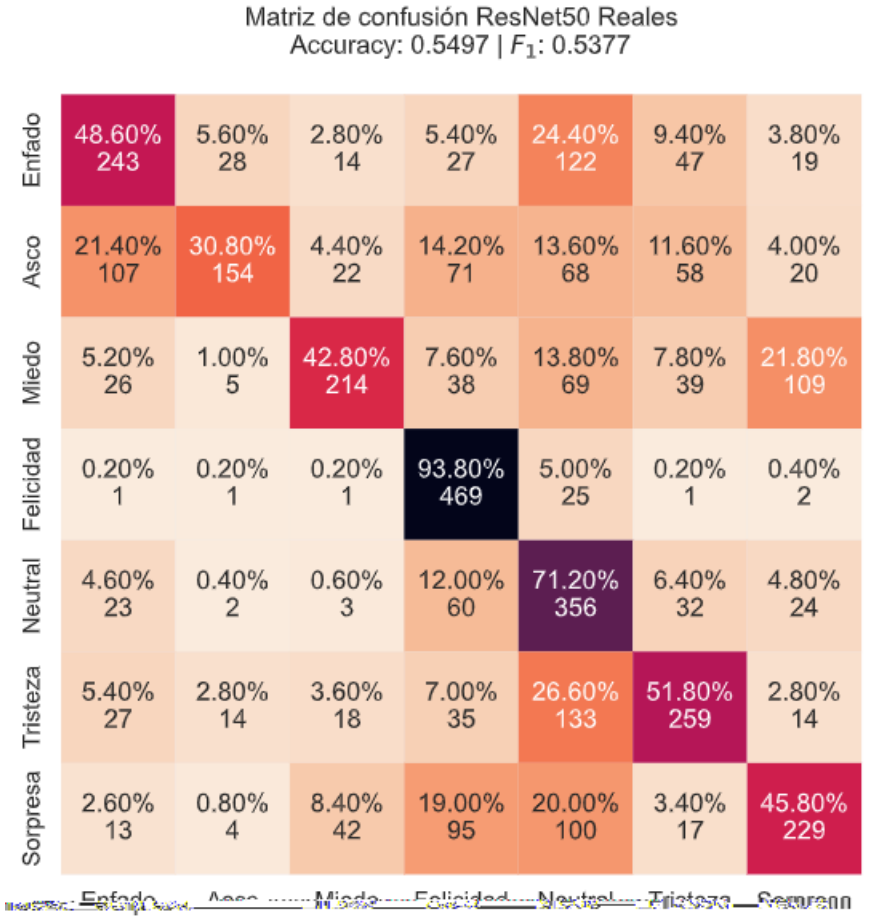
(a) *Dataset* original



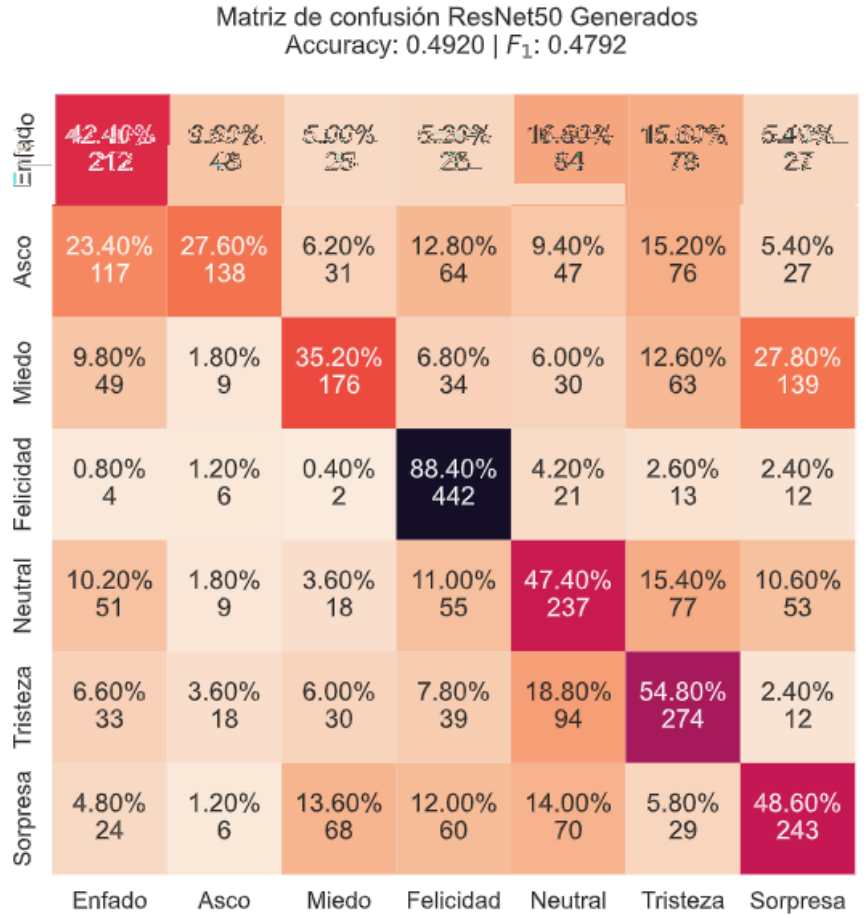
(b) *Dataset* generados

Modelo	Balanceo	Accuracy ↑	Precisión ↑	Recall ↑	F_1 ↑
Datos reales		0.5497	0.5847	0.5497	0.5377
Datos Experimento 2		0.4769	**0.4964**	0.4769	0.4752
	✓	0.4834	0.4924	0.4834	0.4748
Datos Experimento 4		0.4771	0.4903	0.4771	0.4777
	✓	**0.4920**	0.4955	**0.4920**	**0.4792**

05


Matriz de confusión ResNet50 Reales
Accuracy: 0.5497 | F_1: 0.5377
Enfado 48.60% 243 5.60% 28 2.80% 14 5.40% 27 24.40% 122 9.40% 47 3.80% 19
Asco 21.40% 107 30.80% 154 4.40% 22 14.20% 71 13.60% 68 11.60% 58 4.00% 20
Miedo 5.20% 26 1.00% 5 42.80% 214 7.60% 38 13.80% 69 7.80% 39 21.80% 109
Felicidad 0.20% 1 0.20% 1 0.20% 1 93.80% 469 5.00% 25 0.20% 1 0.40% 2
Neutral 4.60% 23 0.40% 2 0.60% 3 12.00% 60 71.20% 356 6.40% 32 4.80% 24
Tristeza 5.40% 27 2.80% 14 3.60% 18 7.00% 35 26.60% 133 51.80% 259 2.80% 14
Sorpresa 2.60% 13 0.80% 4 8.40% 42 19.00% 95 20.00% 100 3.40% 17 45.80% 229
Enfado Asco Miedo Felicidad Neutral Tristeza Sorpresa

(a) M_R


Matriz de confusión ResNet50 Generados
Accuracy: 0.4920 | F_1: 0.4792
Enfado 42.40% 212 9.60% 48 5.00% 25 5.20% 26 16.80% 84 15.60% 78 5.40% 27
Asco 23.40% 117 27.60% 138 6.20% 31 12.80% 64 9.40% 47 15.20% 76 5.40% 27
Miedo 9.80% 49 1.80% 9 35.20% 176 6.80% 34 6.00% 30 12.60% 63 27.80% 139
Felicidad 0.80% 4 1.20% 6 0.40% 2 88.40% 442 4.20% 21 2.60% 13 2.40% 12
Neutral 10.20% 51 1.80% 9 3.60% 18 11.00% 55 47.40% 237 15.40% 77 10.60% 53
Tristeza 6.60% 33 3.60% 18 6.00% 30 7.80% 39 18.80% 94 54.80% 274 2.40% 12
Sorpresa 4.80% 24 1.20% 6 13.60% 68 12.00% 60 14.00% 70 5.80% 29 48.60% 243
Enfado Asco Miedo Felicidad Neutral Tristeza Sorpresa

(b) M_G

06





R

R





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